Roll No. ....

Level 5, 2nd Sem / DVOC (Software Development)

Subject : Data Structure

Time : 2 Hrs. M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

Q.1 The data structure required to check whether an expression contains a balanced parenthesis is?

a) Queue b) Stack

c) Tree d) Array

Q.2 Which of the following tree data structures is not a balanced binary tree?

a) Splay tree b) Red-black tree

c) AVL tree d) B-tree

Q.3 A data structure in which elements can be inserted or deleted at/from both ends but not in the middle is?

a) Priority queue b) Dequeue

c) Circular queue d) Queue

Q.4 What is the value of the postfix expression 6 3 2 4 + _ *?

a) 74 b) 22

c) -18 d) 40

Q.5 Quick sort uses which of the following method to implement sorting?

a) partitioning b) selection

c) exchanging d) merging

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 Queue is a type of______ first out.

Q.7 An______ is a linear collection of data types.

Q.8 Expand DFS.

Q.9 In post order traversal ______ node is traversed first.

Q.10 Directed graph is a type of graph, which________

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 Explain bubble sorting method taking an example of 10 numbers.

Q.12 Differentiate between doubly linked list and circular lists.

Q.13 Write short note on recursion. Give example of the practical application of recursion.

Q.14 Explain collision resolution techniques.

Q.15 Show various flowchart symbols with their meaning.

Q.16 Give the basic operations performed on linked lists.

Q.17 If a =23, b=14 and c=12, then evaluate the postfix expression "ab*c+" to find its value

Q.18 Write down about the steps of insertion sort (descending order) for the list given below:

17, 4,3,5,8,6,11.

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Explain two traversal methods of graphs, taking an example of, at least 10 nodes

Q.20 Define the queue data structure. Write an algorithm to delete and insert an item from the queue and into the queue.